

ODIN Business Plan

Business model and go-to-market strategy for a passive, read-only threat-detection pipeline built for data-diode and air-gapped network monitoring.

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CLASSIFICATION UNRESTRICTED

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00 — EXECUTIVE SUMMARY

A one-way sensor for networks that can't have a way back in.

Data-diode and passive-mirror monitoring is mandated across India's critical infrastructure precisely because it removes the return path an attacker could pivot through — but that same constraint means most commercial NDR/XDR products, built assuming live queries and inline blocking, don't actually fit the deployment. **ODIN** is a detection pipeline designed around that constraint from the ground up: six calibrated ML detectors plus a confidence-gated deep model, a cross-threat correlation engine, and standardized STIX 2.1/CEF export for SOC integration — all built to run purely on what a diode already lets through, with zero assumption of a path back.

ANCHOR CUSTOMER NTRO PS 26145 sponsor, pilot deployment path	ADDRESSABLE SITES, INDIA 180+ NCIIPC-notified critical entities, illustrative	3-YR REVENUE TARGET ₹6.8Cr See §10, base case	HELD-OUT VALIDATION 31k+ flows Confusion-matrix audited, §09
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Figures marked "illustrative" throughout this document are desk estimates built for planning purposes, not third-party market research — flagged as such rather than presented as sourced fact.

The constraint is the point — and the gap in the market

A data diode or passive tap gives a monitoring enclave everything that crosses the wire and nothing to send back. That's deliberate: it closes off the exact pivot path that turns a monitoring system into the next breach. IEC 62443 (industrial control systems), CERT-In's critical-sector directives, and NCIIPC guidance for notified entities all converge on the same requirement — visibility without a return channel.

Most commercial network detection tooling doesn't actually assume that. Darktrace, Vectra, and Cisco Secure Network Analytics are built around continuous model updates pulled from a vendor cloud, live enrichment queries, and often an inline blocking mode — every one of which needs a path the diode is specifically there to remove. Deployments end up either disabling those features (paying for capability they can't use) or routing around the diode entirely (defeating its purpose). Open-source stacks (Zeek/Suricata + a SIEM) close the gap technically but hand the integrator the entire ML/calibration/correlation layer to build themselves.

The gap: a detection pipeline engineered for the one-way constraint as a first-class design requirement, not a mode toggled off — with the ML rigor (calibration, tiered inference, audited metrics) that unidirectional deployments have had to go without.

Six calibrated detectors, one confidence-gated second opinion, one correlation layer

Zeek's passive capture feeds a Kafka event bus; six purpose-built detectors (volumetric DDoS, C2 beaconing, DGA/DNS tunnelling, TLS/QUIC malware via JA3/JA4 and flow metadata, reconnaissance, exfiltration) score every flow, each independently Platt-calibrated so a reported confidence reflects measured precision, not a raw model score. A compact 1D-CNN inspects the raw packet-size/timing sequence for the ~10% of TLS flows that land in a genuinely ambiguous confidence band, refining rather than replacing the fast path. A correlation engine links multi-stage attack chains (recon → C2 → exfiltration) into a single elevated-severity alert. Every alert exports as STIX 2.1 or CEF for a SOC's existing

SIEM/TIP — on request or as a continuously tailed feed — without ever opening a channel back into the monitored network.

Engineering discipline as a selling point

A real end-to-end load-testing pass surfaced and fixed a genuine memory-growth bug in the correlation engine before it reached a live customer — the kind of finding that only shows up under real throughput, not a synthetic unit test. That's the validation posture government procurement scrutiny actually rewards.

Audited, not advertised, accuracy

Per-detector precision/recall/false-positive-rate is published with the exact confusion matrix behind it, including the two detectors that *don't* read 100% — evidence the numbers reflect real held-out evaluation, not a marketing rounding.

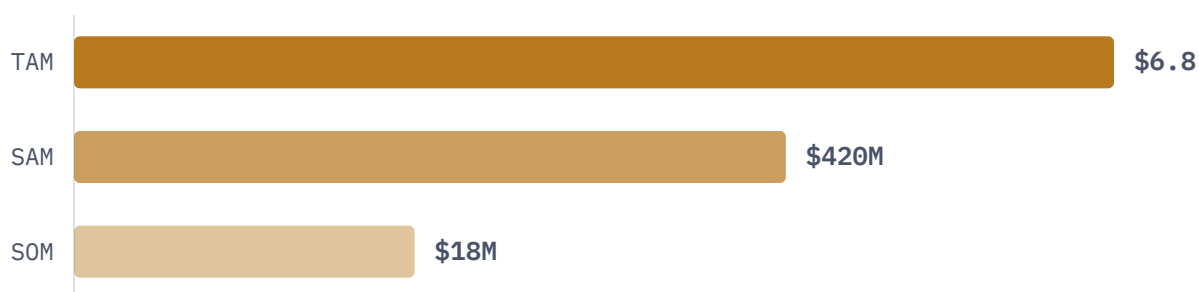
03 — MARKET SIZING

TAM · SAM · SOM

Sized top-down from public industry NDR/OT-security spend estimates, narrowed to the unidirectional/critical-infrastructure segment this product is built for.

Market funnel (USD, illustrative)

TAM: global NDR/OT-security spend, unidirectional-adjacent segment · SAM: India defense + NCIIPC-critical segment · SOM: 3-year realistic capture



TAM \$6.8_B Global NDR/OT-security spend, unidirectional-adjacent, by 2027	SAM \$420_M India defense + NCIIPC critical- sector cyber-monitoring spend	SOM \$18_M Realistic 3-yr capture — ~25–30 sites via pilot + empanelment
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04 — CUSTOMER SEGMENTS

Four tiers, one anchor

TIER	SEGMENT	WHY THEY BUY	TIMELINE
1	NTRO / national CERTs	PS 26145 sponsor; pilot validates against live gateway traffic under real constraints	Yr 1
2	NCIIPC-notified critical infra power, oil & gas, nuclear, rail signaling	Directive-driven diode monitoring mandate; no return-path tolerance	Yr 1–2
3	Regulated BFSI & telecom	RBI/SEBI/DoT cyber-resilience mandates; air-gapped core-network segments	Yr 2–3
4	Allied nations' CERTs	Export once domestically certified & empanelled	Yr 3+

05 — BUSINESS MODEL CANVAS

The full canvas

KEY PARTNERS

- Data-diode / hardware OEMs (Owl Cyber Defense-class, indigenous diode makers)
- CERT-In & STQC (certification, empanelment)
- SI/integrators for govt tendering (L1/GeM routes)
- Academic partners for real attack-traffic capture

KEY ACTIVITIES

- Detector R&D & recalibration against new traffic
- Site-specific baseline tuning at deployment
- CERT-In empanelment & compliance upkeep

- Threat-intel curation (JA3/JA4, DGA families)

KEY RESOURCES

- Calibrated model & training-data pipeline
- STIX/CEF export & correlation engine IP
- Held-out evaluation & benchmark methodology
- Security clearance / empanelment status

VALUE PROPOSITION

- SOC-grade detection with *zero* return path — built for the constraint, not adapted to it
- Calibrated confidence, not raw scores — alert severity means what it says
- Open export (STIX/CEF) into any SOC's existing SIEM/TIP
- Fully auditable metrics & source — indigenous, sovereign-deployable
- Confidence-gated tiering keeps latency bounded under real load

CUSTOMER RELATIONSHIPS

- Direct technical pilot → embedded engineer during onboarding
- AMC-based ongoing relationship, not one-off sale
- Shared threat-intel feedback loop with each site

CHANNELS

- Government tender / GeM listing
- CERT-In empanelled-vendor list
- Direct BD to NCIIPC-notified entities
- SI/integrator partnerships for BFSI/telecom

CUSTOMER SEGMENTS

- National CERTs & defense networks
- NCIIPC-notified critical infrastructure
- Regulated BFSI & telecom operators
- Allied CERTs (export, later)

COST STRUCTURE

- ML/platform engineering payroll (largest line, see §07)
- Compute for training/retraining & benchmark validation
- Certification & compliance (CERT-In, STQC)
- Govt sales cycle & tendering overhead

REVENUE STREAMS

- Per-site license + integration (one-time)
- AMC — retraining, threat-intel updates, support (recurring)
- Managed detection tier for sites without an in-house SOC
- Threat-intel feed subscription

06 — REVENUE STREAMS

Five streams, three already proven at the code level

STREAM	MODEL	INDICATIVE PRICE	STATUS
Per-site license & integration	One-time, tiered by traffic volume	₹35–80L / site	Primary
Annual Maintenance Contract	Recurring, % of license	18% / yr	Recurring
Managed detection tier	Subscription, for sites without SOC staff	₹6–12L / mo	Yr 2+
Threat-intel feed subscription	Subscription — curated JA3/JA4, DGA families	₹2–5L / yr	STIX/CEF export ships today
Professional services	Per-engagement site tuning & custom detectors	₹8–20L / engagement	On-demand

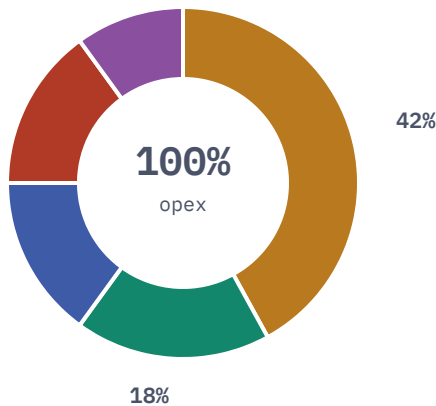
"Status" reflects what's already implemented vs. planned as of this document's revision — the STIX 2.1/CEF export underpinning the threat-intel and managed-detection streams is live in the current codebase, not a roadmap promise.

07 — COST STRUCTURE

Where a rupee of operating cost goes

Annual operating cost breakdown (illustrative, steady state)

Share of total opex once past the pilot phase



■ Engineering & ML ops — 42%
 ■ Compute — 18%
 ■ Sales & tendering — 15%
 ■ Support & AMC delivery — 15%
 ■ Certification & compliance — 10%

- **Engineering & ML ops (42%)** — detector maintenance, recalibration, Tier-2 model retraining as real attack data grows.
- **Compute (18%)** — training/retraining runs and the real end-to-end benchmark harness kept in the loop, not just CI unit tests.
- **Sales & govt tendering (15%)** — GeM/tender cycles run long; budgeted as a fixed overhead, not a variable sales cost.
- **Support & AMC delivery (15%)** — on-site tuning, incident response for customers.
- **Certification & compliance (10%)** — CERT-In empanelment, STQC, periodic re-audit.

08 — GO-TO-MARKET

Pilot → empanelment → sector rollout → export

PHASE	FOCUS	MILESTONE
Yr 1	NTRO pilot deployment against live gateway traffic	Validated throughput number, false-positive rate at a real site
Yr 1–2	CERT-In empanelment; NCIIPC critical-sector rollout	3–5 paid deployments
Yr 2–3	BFISI/telecom expansion; threat-intel subscription launch	15–20 sites; recurring revenue > 30% of total

PHASE	FOCUS	MILESTONE
Yr 3+	Export to allied CERTs	First international deployment

09 — COMPETITION & SWOT

Positioned against incumbents built for a different constraint

VENDOR	BUILT FOR	GAP FOR UNIDIRECTIONAL DEPLOYMENT
Darktrace	Enterprise SOC, cloud-connected	Continuous cloud model updates assume a return path
Vectra AI	Enterprise NDR, active response	Inline/active-response modes unusable behind a diode
Cisco Secure Network Analytics	Enterprise flow analytics	Not calibrated/tiered for diode-scale one-way ingest
Zeek/Suricata + SIEM (DIY)	Open capture layer	No ML calibration, correlation, or export layer out of the box
ODIN	Unidirectional-first, from the ground up	—

Strengths

Indigenous, fully auditable source & models (relevant for classified deployments); calibrated + tiered inference is a real technical moat; STIX/CEF export ships now, not on a roadmap.

Weaknesses

Validation data is still lab/real-tool-heavy rather than years of live production telemetry; small team relative to funded incumbents.

Opportunities

CERT-In's expanding OT-security mandates; Atmanirbhar Bharat procurement preference for indigenous security tooling over foreign vendors for sensitive networks.

Threats

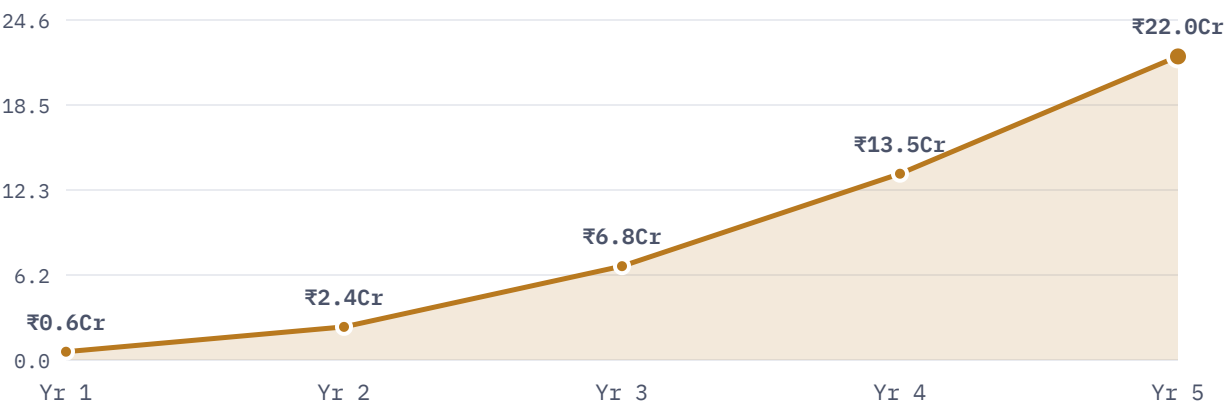
Slow government procurement cycles; well-funded foreign incumbents bundling diode-compatible modes; attacker TTPs evolving faster than any single detector generation.

10 — FINANCIAL PROJECTIONS

Base case, 5 years

Revenue projection (₹ crore, illustrative base case)

Grant-funded pilot in Yr 1; first paid deployments Yr 2; recurring AMC + threat-intel share grows from Yr 3



YEAR	SITES (NEW)	ONE-TIME REVENUE	RECURRING (AMC + INTEL)	TOTAL (₹CR)
Yr 1	1 (pilot)	Grant-funded	–	0.6
Yr 2	4	2.0	0.4	2.4
Yr 3	7	4.4	1.4	6.8
Yr 4	9	7.5	6.0	13.5
Yr 5	10	10.8	11.2	22.0

Breakeven modeled around Yr 2–3 as recurring AMC/threat-intel revenue crosses the fixed engineering + compliance cost base from \$07. Illustrative planning figures, not audited projections.

11 — IMPACT & ROADMAP

National impact, and what ships next

Every additional critical-infrastructure site running real diode-based monitoring is one fewer blind spot in the systems a cyber-physical attack would actually target — power, water, rail signaling, defense networks. An indigenous, fully auditable pipeline also means sensitive deployments aren't sending telemetry patterns through a foreign vendor's cloud-connected model, directly aligned with the National Cyber Security Strategy's sovereignty goals.

Roadmap

HORIZON	WORK
Near	More real-malicious training data (additional CTU-13/Stratosphere scenarios); horizontal scaling toward true multi-gigabit sustained throughput (Flink/Spark, per the architecture's own documented scaling path)
Mid	ICS/SCADA protocol coverage (Modbus/DNP3); multi-site fleet management console; automated STIX threat-intel sharing between agencies
Long	5G-core/QUIC-heavy encrypted traffic coverage; hardened export-ready appliance for allied-CERT deployment

12 — RISKS

What could actually go wrong

Validation-data risk

Several detectors are still trained on lab/synthetic data rather than years of live traffic. Mitigated by publishing exact confusion matrices per detector (not just headline scores) and treating each new real capture as a documented, honest update — not a re-marketed number.

Procurement-cycle risk

Government tendering timelines can stretch a funded pilot into a multi-year sales cycle. Mitigated by the AMC-first revenue model, which doesn't depend on a large upfront license to sustain the business between tender cycles.

Throughput-scaling risk

Current sustained throughput is validated on a single-node setup, not yet a distributed deployment. The architecture's detector/dispatch boundary is already designed to swap in Flink/Spark without touching detection logic — a scaling path, not a rewrite.

Operational-robustness risk

A real load-testing pass already found and fixed a memory-growth bug in the correlation engine before any customer hit it — the process that catches this class of issue is now a standing part of the release discipline, not a one-off.